

University of Salerno

MSc in Computer Engineering, DIEM

Solar Filament Segmentation

Machine Learning project work

Team

2026_MLrec_gr02

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Academic year 2025/2026

1 The task, the data and the metrics

Solar filaments are ribbons of cooler plasma suspended over the solar disk, visible as dark, elongated structures in H-alpha images. The task is binary segmentation with an instance twist: given a 2048×2048 observation, predict which pixels belong to a filament, split the mask into one connected component per filament, and submit each component as a COCO run-length encoding.

1.1 What the dataset looks like

The training set contains **707 observations** from the six stations of the GONG network (Big Bear, Cerro Tololo, Learmonth, Mauna Loa, El Teide, Udaipur), taken between 2011 and 2022 with a strong skew towards the 2011–2015 solar maximum. The annotations come from the MAGFiLO catalogue and carry a structure that shaped the whole project: each of the **1154 annotation records** is *one annotator's view* of one observation. 411 files were annotated by a single expert, 145 by two and 151 by three, and the annotators genuinely disagree – the mean Dice between two annotators of the *same* image is 0.62 (median 0.65).

The 8199 annotated filaments are small and thin: the median filament covers about 1230 pixels at native resolution (interquartile range 670–2430, 15% below 500), and all filaments together cover only $\approx 0.4\%$ of the pixels. Three consequences drive the design. The extreme class imbalance rules out a plain cross-entropy loss; structures a few pixels wide are destroyed by aggressive downscaling, which will make *resolution* the most productive variable of the study; and with only 494 training observations (the training set is explicitly not extendable), transfer learning and augmentation are the only remaining levers.

1.2 Metrics: IoU, Dice, and Panoptic Quality

Two masks P (prediction) and G (ground truth) can be compared by their **Intersection over Union** or by the closely related **Dice coefficient**:

$$\text{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}, \quad \text{Dice}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \quad (1)$$

Both are pixel-level measures. Dice is what the course grades; the Kaggle submission, however, asks for one row per *filament*, and the leaderboard reflects **Panoptic Quality**. Predicted and annotated instances are matched when their IoU exceeds 0.5 (matched pairs are true positives, unmatched predictions false positives, unmatched annotations false negatives), and

$$\text{PQ} = \frac{\sum_{(p,g) \in \text{TP}} \text{IoU}(p, g)}{|\text{TP}| + \frac{1}{2}|\text{FP}| + \frac{1}{2}|\text{FN}|} = \underbrace{\frac{\sum \text{IoU}}{|\text{TP}|}}_{\text{SQ}} \times \underbrace{\frac{|\text{TP}|}{|\text{TP}| + \frac{1}{2}|\text{FP}| + \frac{1}{2}|\text{FN}|}}_{\text{RQ}}. \quad (2)$$

The factorisation matters: *Segmentation Quality* measures how well matched instances overlap, *Recognition Quality* measures how many instances are found at all. The gap between Dice and PQ is the whole story of this project – a filament broken into three fragments, or a correct mask surrounded by twenty ten-pixel specks, moves Dice by a fraction of a percent and collapses RQ.

The competition's own PQ, published in the organisers' self-evaluation notebook, differs from the textbook definition in three ways that all change what a good prediction looks like: ground truth is *one instance per annotation*, not connected components of the union of all annotators' polygons; *each annotator is a separate evaluation entry*, so one prediction is scored two or three times against experts who disagree; and TP/FP/FN accumulate *globally* over the split rather than per image. The pipeline re-implements exactly this definition and verifies it against the

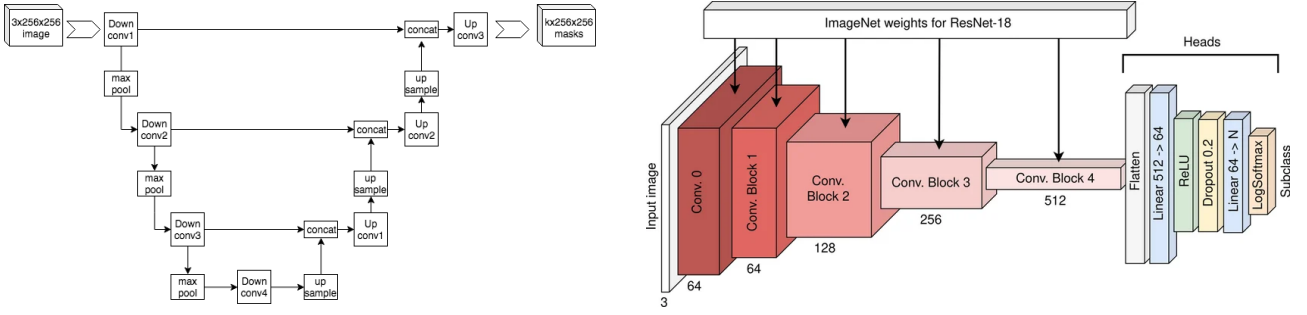


Figure 1: Left: the U-Net encoder–decoder with skip connections. Right: the ResNet18 encoder that replaces the from-scratch one from Experiment 2 onwards; its classification head (right of the figure) is discarded and the four convolutional blocks feed the U-Net decoder through the skip connections.

reference code (agreement to 3×10^{-8}), so every validation number in this report is directly comparable with the leaderboard.

1.3 How good is good enough

The inter-annotator Dice of 0.62 is a practical ceiling: past it, a model is not segmenting better, it is fitting one expert’s notion of where a filament ends. The number reframes the problem – from the second experiment onwards the model is already at that ceiling on Dice while PQ sits around 0.25, so the headroom is entirely in how the mask is *organised into instances*. Every hypothesis that follows is built on that observation.

2 Model and pipeline

2.1 A U-Net with a pre-trained ResNet18 encoder

A classification CNN compresses an image into one prediction; segmentation needs one per pixel, so the spatial detail the encoder discards has to be put back. U-Net does this with a symmetric encoder–decoder and **skip connections** that copy each encoder stage to the decoder stage of the same resolution (Figure 1, left). For thin, elongated, irregular structures the skips are not a refinement – they are what keeps a filament from dissolving in the bottleneck. The network emits one logit per pixel; a sigmoid turns it into a probability, and a threshold selected on validation into the binary mask.

The baseline (Experiment 1) uses a from-scratch encoder with 1.9M parameters. From Experiment 2 onwards the encoder is a **ResNet18 pre-trained on ImageNet** (Figure 1, right, without its classification head), fine-tuned end to end: with fewer than 500 training observations, learning basic edges and textures from scratch wastes the data budget. The decoder stays the same, so the comparison isolates the encoder. The architecture is then held fixed for the rest of the study on purpose – one variable per experiment is what makes the comparisons mean anything.

One free optimisation is worth a paragraph. Feeding grayscale to an ImageNet encoder normally means replicating the channel three times and normalising with the RGB mean m_c and standard deviation s_c . Writing out the stem pre-activation for a grayscale image g shows the redundancy:

$$\sum_c W_c * \frac{g - m_c}{s_c} = \left(\sum_c \frac{W_c}{s_c} \right) * g - k, \quad (3)$$

where k is a per-output-channel constant that the batch norm immediately after absorbs into its running mean. The three-channel stem is therefore replaced by an exactly equivalent *single-*

channel one: the pre-trained weights are fully preserved, the input tensor shrinks threefold, and normalisation disappears from the data pipeline.

2.2 Loss and training protocol

The loss is an equal mix of binary cross-entropy and soft Dice. Each term does a job the other cannot: Dice is immune to the 0.4% positive rate but says nothing about confidence, while BCE keeps the probabilities *calibrated*, which is what makes the threshold search of Section 4 well behaved. Experiment 3 shows what happens when that calibration is lost.

The protocol is identical in every run. The split is on **file names**, never on annotation records – the same JPEG appears up to three times in the COCO file, once per annotator, and splitting on records would put one copy in training and another in validation, inflating every score. Seed 22 gives 494/106/107 observations (806/169/179 records), written to `splits/split_seed22.csv`; the internal test split is consulted once, after everything is frozen. Augmentation uses the eight symmetries of the square (filaments have no canonical orientation) plus photometric jitter. Optimisation is AdamW, batch size 2, at most 20 epochs, early stopping on validation Dice with patience 8. The checkpoint is selected on **Dice**, not PQ: PQ depends strongly on the threshold and post-processing, tuned afterwards on the same validation set, and selecting weights with it would optimise two coupled quantities at once.

2.3 Engineering for iteration speed and memory

Decoding a 2048^2 JPEG and rasterising its polygons costs more than a forward and backward pass; done inside the dataset, that price recurs every epoch. All images and masks are therefore decoded *once* into memory-mapped uint8 arrays (masks bit-packed: the whole ground truth at 1024^2 takes 150 MB instead of 1.2 GB), and every run reuses the cache. Training runs in mixed precision with gradient checkpointing at the higher resolutions; peak GPU memory is measured at the moment a step actually peaks, not estimated from parameter counts. Against the assignment’s optional budget (5 GB training, 4 GB test), inference stays well inside at 2.5 GB; training measured 3.7 GB at 1024^2 and 5.2 GB for the full-frame 1536^2 recipe of the final model – a windowed-native configuration (Experiment 13, Section 5) brings the same step to 3.0 GB, though it lost PQ and was not kept. Validation probability maps are stored as uint8 – the threshold grid steps far coarser than $1/255$ – which is what makes the joint operating-point search of Section 4 affordable at native resolution.

3 First campaign: experiments 1–5

The experiments are not a chronology. Before training anything, the observations of Section 1 were turned into an ordered list of hypotheses, each changing exactly one thing and each falsifiable by a single number: (1) a plain encoder–decoder already finds the main filaments; (2) with fewer than 500 observations the encoder is the bottleneck, so ImageNet features should beat random initialisation; (3) Dice is blind to fragmentation, so a topology-aware loss should raise PQ; (4) part of the residual is the H-alpha *domain* rather than the loss; (5) what limits PQ is *scale*. Hypotheses 3 and 5 make opposite predictions about where the error lives, and only one survives contact with the data.

Experiments 1–2, baseline and transfer learning. The from-scratch U-Net reaches Dice 0.6415 at the last epoch, still underfitted, with its best threshold at the edge of the grid – a sign of poorly calibrated probabilities that points at the encoder. Replacing it with the pre-trained ResNet18 lifts Dice to 0.6552 at epoch 8 and, more usefully, flattens the threshold–Dice curve: for a submission whose threshold must be frozen blind, that calibration is worth more than the point of Dice.

Error analysis. At 0.655 Dice is already at inter-annotator level, yet PQ is 0.2505. Decomposing it: $SQ = 0.61$ (matched instances overlap fine) but $RQ = 0.38$. The false positives are a swarm of tiny spurious components (median 7 pixels), not fragmentation – 74% of ground-truth filaments over 32 pixels are predicted as a single component. This diagnosis is what makes Experiment 3 worth running, and what post-processing later exploits.

Experiment 3, a topology-aware loss – falsified. Replacing half of the loss with Tversky ($\beta = 0.7$) plus clDice (a Dice between soft skeletons) collapses validation Dice to 0.4937 and PQ to 0.0618, while the training loss decreases: the model optimises the wrong thing successfully. Two lessons. Tversky’s recall bias thickens everything and destroys the calibration transfer learning had bought; and at 512^2 many filaments are one or two pixels wide, so the soft skeleton is noise on exactly the structures clDice was meant to protect – a scale problem wearing the costume of a loss problem, and the first hint that hypothesis 5 aims at the right variable.

Experiment 4, the H-alpha domain. ImageNet normalisation (folded into the stem), photometric jitter and cosine annealing: Dice 0.6591, PQ 0.2656, and the validation curve now improves to the last epoch instead of stalling at 8.

Experiment 5, resolution – the largest single gain. Nothing changes except the input size, $512 \rightarrow 1024$. Dice gains one point (0.6701) while PQ jumps from 0.2656 to 0.3397, +28% relative (Figure 2). The asymmetry is the point: doubling the resolution barely changes how much of the disk is covered correctly, and changes a great deal about how many filaments exist as recognisable objects. What looked like a loss problem was a sampling problem.

Table 1: First campaign. Dice and PQ at threshold 0.5 without post-processing, so the columns are directly comparable; the operating point is selected afterwards (Section 4).

Exp.	Hypothesis under test	Dice	PQ	Best epoch
1	U-Net from scratch (baseline)	0.6415	–	20
2	ImageNet ResNet18 encoder	0.6552	0.2505	8
3	Tversky + clDice loss	0.4937	0.0618	13
4	Photometric aug. + normalisation + cosine	0.6591	0.2656	18
5	Resolution $512 \rightarrow 1024$	0.6701	0.3397	15
<i>inter-annotator agreement</i>		<i>0.62</i>	–	

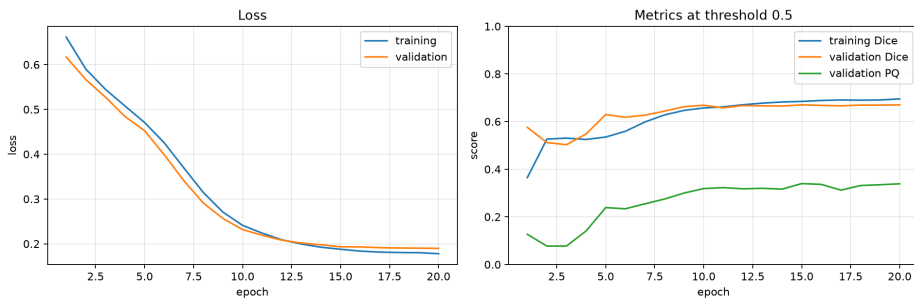


Figure 2: Experiment 5 at 1024×1024 . Validation PQ, the lowest curve on the right, gains far more from the change of scale than Dice does.

4 From probabilities to a submission

Training produces a probability map; four decisions turn it into the mask that is scored, and together they are worth more PQ than any single training change. All are selected on validation

only, and *jointly*, because they interact – tuning the threshold first and the filters afterwards locks in a worse point (on Experiment 5, 0.389 sequential against 0.401 joint).

- the **threshold** that binarises the probability map;
- **morphological closing**, which reconnects a filament the network split in two, converting two FP and one FN into one TP;
- a **minimum component size**, deleting the tiny spurious components of Section 3 that each cost a full false positive;
- a **confidence gate**: a component survives only if its probability peaks above the gate somewhere. The threshold alone does two jobs at once – deciding *which* blobs exist and *how fat* they are – so raising it to kill weak blobs also thins every real filament. The gate takes over the existence decision and leaves the outlines where the threshold drew them.

Test-time augmentation over the eight square symmetries averages eight probability maps and is kept behind a flag; the threshold must then be re-selected on the averaged maps, otherwise TTA looks harmful when it is not.

4.1 Scoring what the leaderboard scores, where it scores it

Two things had to be right for a validation number to mean anything. **What**: the official PQ of Section 1.2, re-implemented and verified to 3×10^{-8} against the organisers’ code. On the union of two or three annotators an image carries about eleven filaments against under seven for any single one, so a model tuned against the wrong definition systematically over-predicts. **Where**: the submission is written at 2048², but early operating points were selected at training resolution – a protocol mismatch, since a minimum size tuned at 1024 means something else at 2048 and the number of connected components is not preserved by resizing a mask at all. The fix is to enlarge the *probability* map to native resolution and only then threshold, post-process and score. Measured on Experiment 5, selecting at native resolution against the official metric is worth +0.020 Dice and +0.006 PQ over the recipe that would otherwise have shipped (Table 2) – for free, with no retraining.

Table 2: Experiment 5, same checkpoint, four ways of choosing the operating point, all scored with the competition’s own metric at 2048².

Selection protocol	t	closing / min. size	Dice	PQ	SQ	RQ
tuned at 1024 on PQ	0.65	1 / 512	0.6462	0.3901	0.660	0.591
tuned at 1024 on Dice+PQ	0.90	1 / 256	0.6590	0.3964	0.662	0.599
tuned at 2048, internal metric	0.70	2 / 256	0.6653	0.3965	0.660	0.601
tuned at 2048, official metric	0.75	2 / 256	0.6665	0.4022	0.662	0.608

A last refinement came late and cost nothing: the leaderboard reports PQ and nothing else, but the recipe had historically been frozen on the *balanced* objective Dice + PQ. Re-selecting the final model’s point on pure PQ, with a second, finer threshold pass around the coarse optimum (Figure 3), moves it from 0.4194 to **0.4264** (threshold 0.48, closing 1, minimum size 384, gate 0.97). Cross-selection – choosing the point on one half of the validation files and scoring it on the other – puts the overfitting cost of the finer grid at only 0.0024, with both halves independently selecting the same threshold and gate: the refinement is real, not grid noise.

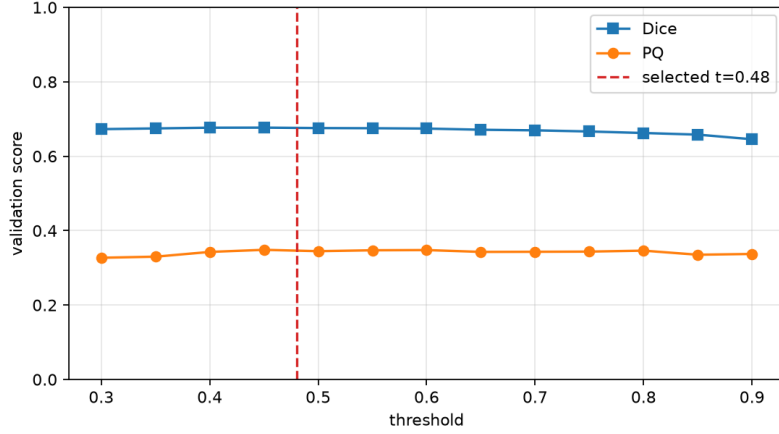


Figure 3: Validation Dice and PQ against the threshold for the final model (no post-processing). The two metrics peak at different points; the shipped recipe is selected on PQ, jointly with closing, minimum size and gate.

5 Second campaign: instances

The first campaign ended with a diagnosis – what limits the score is instance recognition – and the second attacked it, keeping the same discipline: one hypothesis per run, falsified results reported with the same care as confirmed ones. From Experiment 7 on, the internal test split is folded into training (+22% data), legitimate exactly because it is never used to choose anything.

Table 3: Second campaign, each run at its own operating point, scored with the official metric at 2048² (not comparable with Table 1, which is at fixed threshold without post-processing).

Exp.	What changes	Dice	PQ	Verdict
7	test split folded in, still 1024	0.6544	0.3775	superseded by 8
8	1024 → 1536, D4 TTA, confidence gate	0.6729	0.4194	final model
9	+ cDice term at 1536, warm start	0.6686	0.4064	falsified
10	±25% scale jitter	0.6670	0.4003	falsified
11	detect-then-refine cascade	0.6560	0.3947	falsified
12	soft consensus targets	0.6687	0.3863	falsified
13	native 2048, random 1536 windows	0.6692	0.4066	falsified
14	instance-balanced Dice, weight 0.35	0.6572	0.3611	falsified
15	BCE positive-class weight 20	0.6769	0.4198	falsified by 0.007

5.1 The ceiling is annotator disagreement, and it binds SQ

Factorised, the shipped point’s $PQ = 0.4264$ is $SQ = 0.668 \times RQ = 0.638$ over 1144 ground-truth instances (TP 712, FP 376, FN 432). Across every capacity, resolution and architecture tried in thirteen experiments, SQ moved inside 0.66–0.68 and no further – against a matched-pair agreement between human annotators of 0.59. The conclusion is uncomfortable but useful: **SQ is a property of the annotation process, not of the model**. Every point of PQ still on the table goes through RQ: finding more of the filaments annotators marked, predicting fewer they did not.

5.2 Where the error actually is

At the shipped operating point, the 432 false negatives split into 244 *pure misses* (not one predicted pixel on the instance), 159 *near misses* (a blob is there but stays at $\text{IoU} \leq 0.5$,

median predicted/true area 0.62 – the model systematically under-segments) and 29 *splits*; the 376 false positives into 128 *hallucinations*, 239 on annotated but unmatched filaments, and 9 *merges*. Instance recall falls sharply with size, from 0.34 below 500 pixels to 0.75 above 5000 (Figure 4); at the recall-leaning balanced point the smallest bucket rises to 0.475 at the cost of 62 extra false positives, and that is where the diagnosis behind the next campaign was read. Two families dominate – small-instance recall and under-segmentation; split/merge errors are under 4% of the total, which is why no split/merge machinery was ever built.

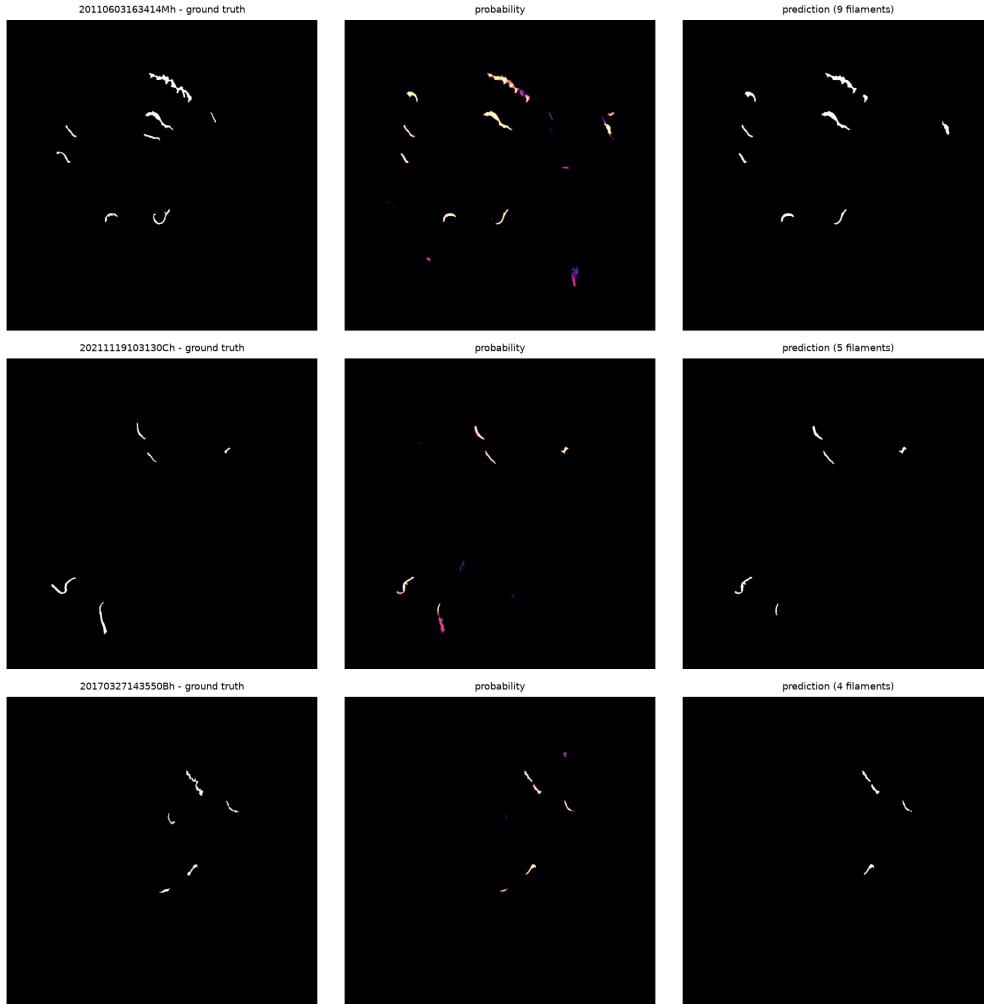


Figure 4: Final model on three validation observations: annotator ground truth, predicted probability map, and instances at the shipped operating point. The errors are concentrated in small, faint filaments.

Two caveats qualify these counts, and both were checked. First, the organisers have confirmed in the competition forum that the annotations are *incomplete* – visible filaments may be missing – so part of the 128 hallucinations are plausibly real filaments no annotator marked; measured on the multi-annotator files, even blobs whose probability peaks above 0.85 are marked by only 63% of annotators on average, which is why the confidence gate, not a recall push, sits at the PQ optimum. Second, the judging committee reads the *many-to-one* / *one-to-many* distribution alongside the score (Figure 5): here the matching structure is nearly one-to-one – of 1144 annotated instances, 900 are touched by at least one prediction and only 67 by more than one; of 1088 predictions, 960 touch an annotated filament and only 20 touch more than one.

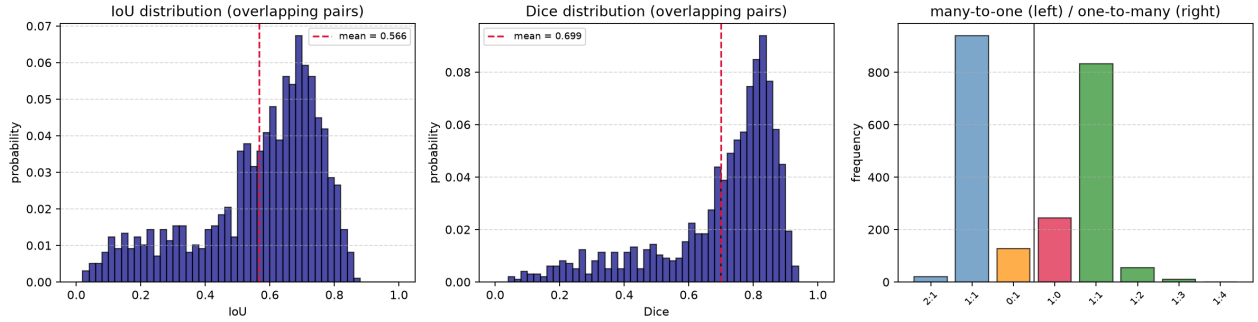


Figure 5: The distributions the organisers’ self-evaluation notebook reports, for the shipped model on validation: IoU and Dice over every overlapping ground-truth/prediction pair, and the many-to-one / one-to-many relations (a hit is any overlap; the metric’s IoU > 0.5 gate is stricter).

5.3 Mechanisms measured and rejected

Several plausible mechanisms were evaluated and discarded – most of them cheaply, on the *frozen* probability maps of the final model, reproduced bit-exactly on a second machine so that any instantiation idea could be measured in minutes without a training run.

Topology and cascade machinery. cDice at 1536, where the soft skeleton is finally meaningful, still loses 0.013 PQ (Exp. 9); a detect-then-refine cascade loses 0.025 (Exp. 11). Both attack fragmentation, which the error analysis had already shown to be marginal. **Soft consensus targets** (Exp. 12) turn annotator disagreement into the label; calibration barely moves and PQ drops 0.033. **Scale jitter** (Exp. 10) costs 0.019. **Per-component local thresholds** ($\tau_c = \alpha \cdot \text{peak}_c$, or a quantile of the component’s own probability mass) should fix under-segmentation without over-fattening correct components; measured, they lose to the plain global threshold, because every surviving component peaks near 1.0 and the local cut collapses into a global one. **Per-component dilation** (nearest-component distance transform, so merges are impossible) loses at every radius – $\delta = 1$ px already costs 0.03 PQ – because near misses are not eroded copies of the truth but differently shaped, typically missing a faint tail. **Ensembling probability maps** loses twice: with the weaker siblings (Exps. 9, 10, 12), whose miscalibration dilutes the marginal detections, and even with the near-equal, oppositely-biased Exp. 15 – a +0.003 aggregate gain that scores *below* the single model under cross-selection. The emerging rule: on 169 validation entries, any gain below $\approx +0.005$ cannot be told from selection noise. And **Mask R-CNN** was rejected on paper: it rebuilds each mask from a 28×28 grid inside an axis-aligned box, an upsampling of $\sim 25\times$ per axis for structures whose defining property is being a few pixels wide and hundreds long.

Post-hoc merging of split filaments was bounded and then falsified. An oracle that merges, with ground-truth knowledge, exactly the prediction pieces that would flip a split into a match gains +0.017 PQ – but only 18 of its 29 merge groups flip cleanly, because the annotators disagree on connectivity itself. No blind mechanism captures that ceiling: relabelling two instances as one when they are connected at a low probability threshold (no pixel added, so Dice is untouched) gains +0.004 in aggregate, yet under cross-selection the two validation halves pick opposite ends of the parameter grid and the transferred gain is -0.006 – selection noise, not signal; merging by spatial proximity loses at every useful radius. Low-probability bridges cannot tell one broken filament from two nearby ones.

The extent and merge mechanisms failing identically is the finding of the campaign: the under-segmentation and the fragmentation both live in the model’s probability support, not in the thresholding. No post-processing can claim them; only training changes can.

5.4 Validation versus leaderboard

The public score is systematically below validation, and the gap decomposes into measured parts: an early submission was produced by a non-final recipe (worth ~ 0.02); ordinary generalisation to a different annotator draw (~ 0.026 , measured once on the internal test split); and composition (≤ 0.01 – PQ stratified by annotator multiplicity, GONG station and year varies within counting noise, and the test set’s station/year mix matches validation). The submission path itself was audited clean: disjoint components by construction, RLE conventions matching the organisers’ scorer, zero components off the solar disk.

6 Results, conclusions and what comes next

The delivered model is Experiment 8 – ResNet18-U-Net at 1536^2 , D4 test-time augmentation – with the operating point re-selected on pure PQ: validation Dice 0.668, PQ **0.4264** ($t = 0.48$, closing 1, minimum size 384, confidence gate 0.97). The submission is fully reproducible: the training notebook re-derives exactly this recipe from the frozen checkpoint, and the inference notebook regenerates a byte-identical `submission.csv`.

Three lessons generalise beyond this dataset. **Measure what the score measures, where it measures it**: selecting the operating point at native resolution, against the organisers’ own PQ definition, and on the objective the leaderboard actually reports was worth $+0.020$ Dice and $+0.031$ PQ combined – all free, with no retraining. **Pick the metric that sees the failure mode**: Dice crossed the inter-annotator ceiling at Experiment 2 and stayed flat; every real improvement afterwards was visible only in PQ’s recognition term. An evaluation restricted to the graded metric would have declared the project finished ten experiments early. **A falsified hypothesis is not a wasted run**: ten ideas were measured and rejected in this study, and each rejection redirected the effort – Experiment 3’s failure exposed the resolution problem, and the failure of every post-hoc extent mechanism (Section 5.3) is precisely what shows the remaining error must be attacked in training.

One of the prepared follow-ups has already been run and falsified: training on random 1536^2 windows of the native 2048^2 frame (Experiment 13) lost 0.020 PQ and pushed recall below 500 pixels down to 0.31 – windowed training gives up the full-disk context without buying the small filaments, and it closes the resolution road ($+0.136$, $+0.009$, -0.020 across the three scale steps). The instance-balanced loss (Experiment 14) was then falsified as a *dominant* term: with the global Dice demoted, false-positive control collapsed and small-instance recall fell to 0.09 – yet recall on 2000–5000 pixel instances improved to 0.76, so the term earns a retest at nudge weight, the same arc as Experiments 3 and 9. The positive-class weight (Experiment 15) then delivered the campaign’s most instructive miss: the under-segmentation vanished (near-miss area ratio $0.62 \rightarrow 0.95$), pure misses fell by a fifth, and Dice set the project record at 0.677 – but the fatter predictions inflated false positives in parallel, landing 0.007 under the champion. What closes the campaign is a statistical fact: on 169 validation entries, any gain below $\approx +0.005$ is indistinguishable from selection noise. The one prepared step that attacks exactly that is *five-fold cross-validation*: every observation trains a model, and the out-of-fold predictions support an operating-point search on $6.7\times$ more annotator entries – turning the gains this campaign could only suspect into ones it can measure.